# DIGITAL LOGIC FUNDAMENTALS

| | |
|---|---|
| Course Code | :30801220 |
| Credit Hours | :3 |
| Prerequisite | : 30801101 |

# Chapter 6

# THE KARNAUGH MAP

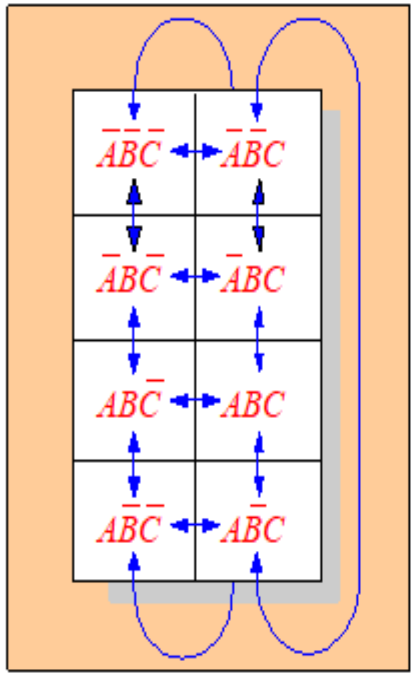
| AB \ CD | 00 | 01 | 11 | 10 |
|---|---|---|---|---|
| 00 | 1 | 0 | 1 | 1 |
| 01 | 0 | 1 | 1 | 1 |
| 11 | 0 | 0 | 1 | 1 |
| 10 | 1 | 0 | 1 | 1 |

# K-Map

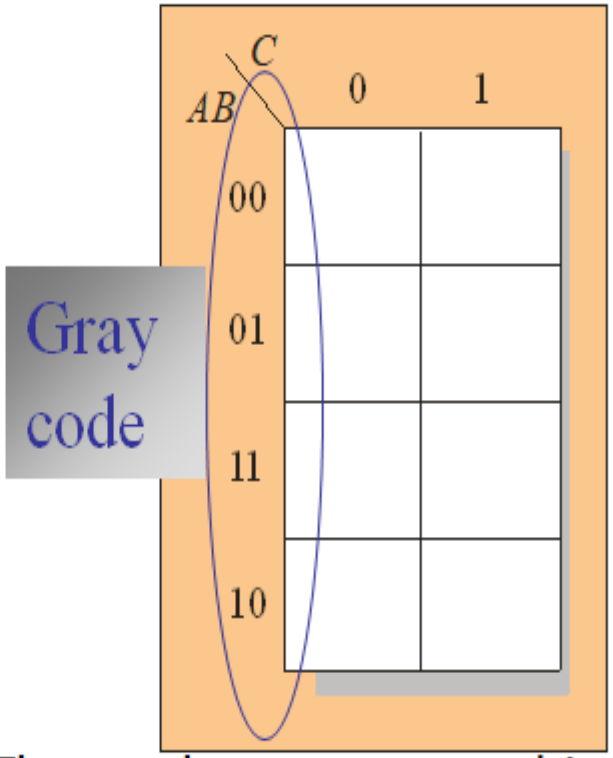
- The Karnaugh map is an array of cells
- Each cell represents a binary value of the input variables.
- Can facilitate to produce the simplest SOP or POS expression
- The number of cells is $2^n$, $n$ is number of variables

# K-Map (cont.)

3 variables so 8 cells

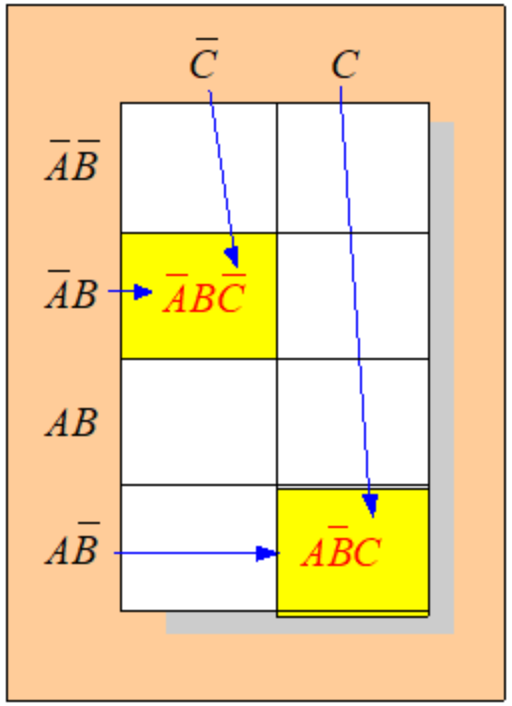



Each cell differs from an adjacent cell by only one variable




The numbers are entered in gray code, to force adjacent cells to be different by only one variable.

# Kmap for 3 variables




How to read !!

| AB \ C | 0 | 1 |
|---|---|---|
| 00 | $\bar{A}\bar{B}\bar{C}$ | $\bar{A}\bar{B}C$ |
| 01 | $\bar{A}B\bar{C}$ | $\bar{A}BC$ |
| 11 | $AB\bar{C}$ | $ABC$ |
| 10 | $A\bar{B}\bar{C}$ | $A\bar{B}C$ |

Full representation for 3 variables.
(in fact it is n-Dimension truth table)

# Karnaugh Map SOP Minimization

**Aim:** You should be able to utilise K-Map to simplify Boolean expression to their minimum form.

## Convert SSOP to K-Map

What we know is each product term in SSOP relates to "1" in corresponding truth table, but K-Map is, in deed, a form of n-dims truth table.

Hence the way to convert SSOP to K-map is similar to the way to convert SSOP to truth table !!

$\bar{A}\bar{B}\bar{C}+\bar{A}\bar{B}C+A\bar{B}\bar{C}+A\bar{B}\bar{C}$

| AB \ C | 0 | 1 |
|---|---|---|
| 00 | 1 | 1 |
| 01 | | |
| 11 | 1 | |
| 10 | 1 | |

# Karnaugh Map SOP Minimization (cont.)

## Convert non-standard SOP to K-Map

Assume that the domain of Boolean is {A,B,C} and the expression we consider is A

convert A (non-SOP) to SSOP as follows:-

$$A = A(B+\bar{B})(C+\bar{C})$$
$$= (AB+A\bar{B})(C+\bar{C})$$
$$= ABC+AB\bar{C}+A\bar{B}C+A\bar{B}\bar{C}$$

*ABC'

| AB \ C | 0 | 1 |
|---|---|---|
| 00 | | |
| 01 | | |
| 11 | 1 | 1 |
| 10 | 1 | 1 |

Observe that A is related to all cells related to the binary "1" of A

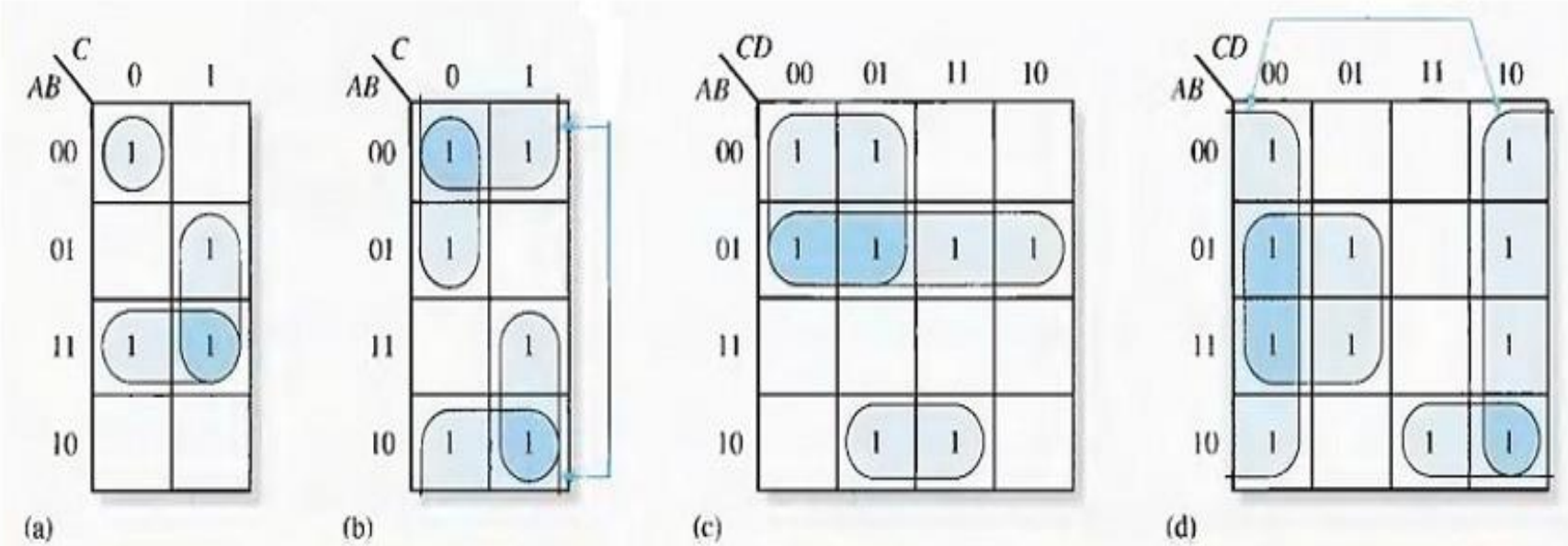
# 4 Variables K-Map

| AB \ CD | 00 | 01 | 11 | 10 |
|---|---|---|---|---|
| 00 | 0 | 1 | 3 | 2 |
| 01 | 4 | 5 | 7 | 6 |
| 11 | 12 | 13 | 15 | 14 |
| 10 | 8 | 9 | 11 | 10 |

## Minterm representation

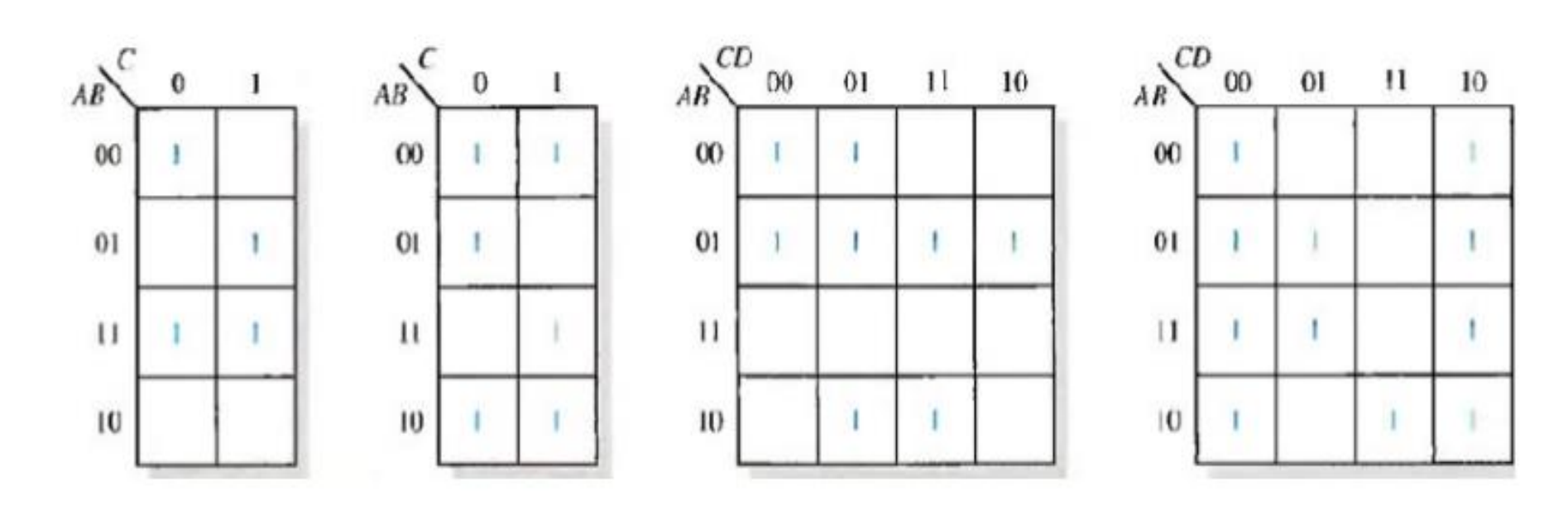
| AB \ CD | 00 | 01 | 11 | 10 |
|---|---|---|---|---|
| 00 | A'B'C'D' | A'B'C'D | A'B'CD | A'B'CD' |
| 01 | A'BC'D' | A'BC'S | A'BCD | A'BCD' |
| 11 | ABC'D' | ABC'D | ABCD | ABCD' |
| 10 | AB'C'D' | AB'C'D | AB'CD | AB'CD' |

# Grouping 1s examples

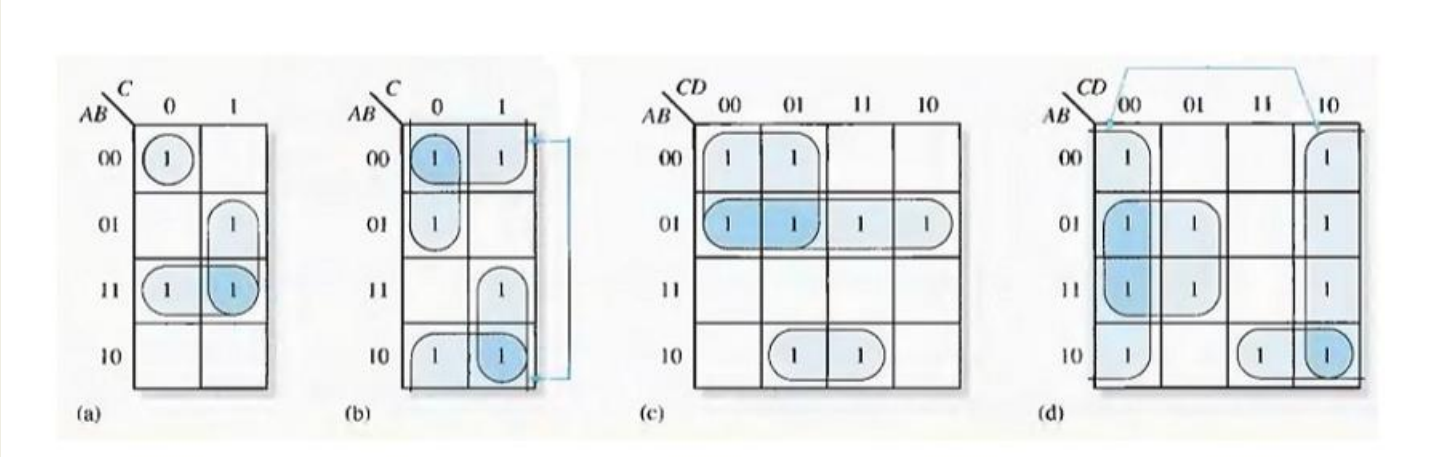


# Exercise

- Group the 1s in each of the Karnaugh maps in the following figures:

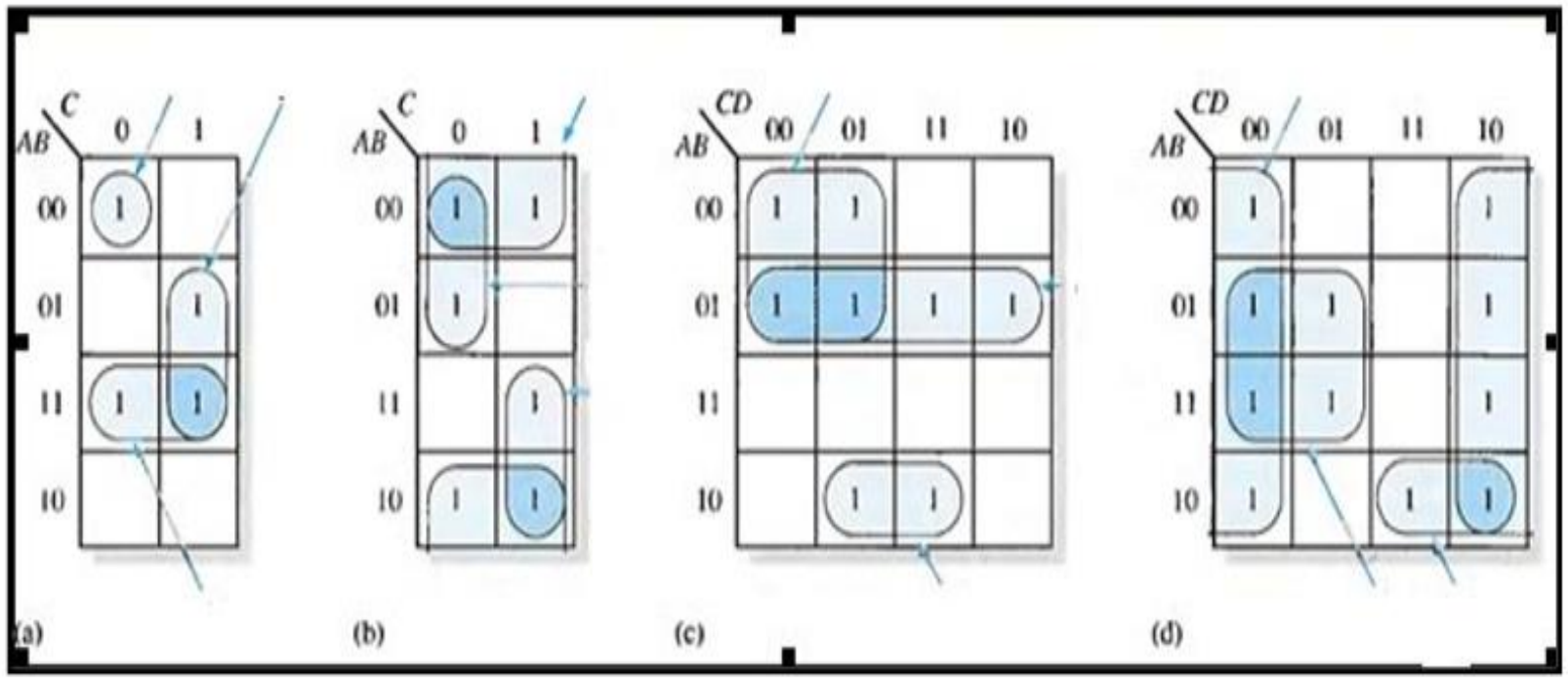


# Solution



# Exercise

- Determine the product terms for each of the Karnaugh maps in the following figures and write the resulting minimum SOP expression



# Solution

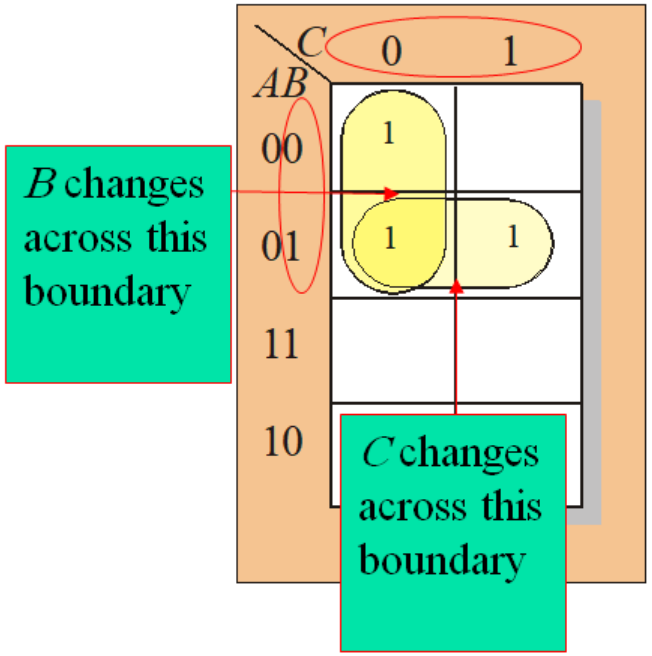
(a) $AB+BC+\overline{A}\,\overline{B}\,\overline{C}$

(C) $\overline{A}B + \overline{A}\,\overline{C} + A\overline{B}D$

(b) $\overline{B} + A\,C + \overline{A}\,\overline{C}$

(d) $\overline{D} + A\overline{B}C + B\overline{C}$

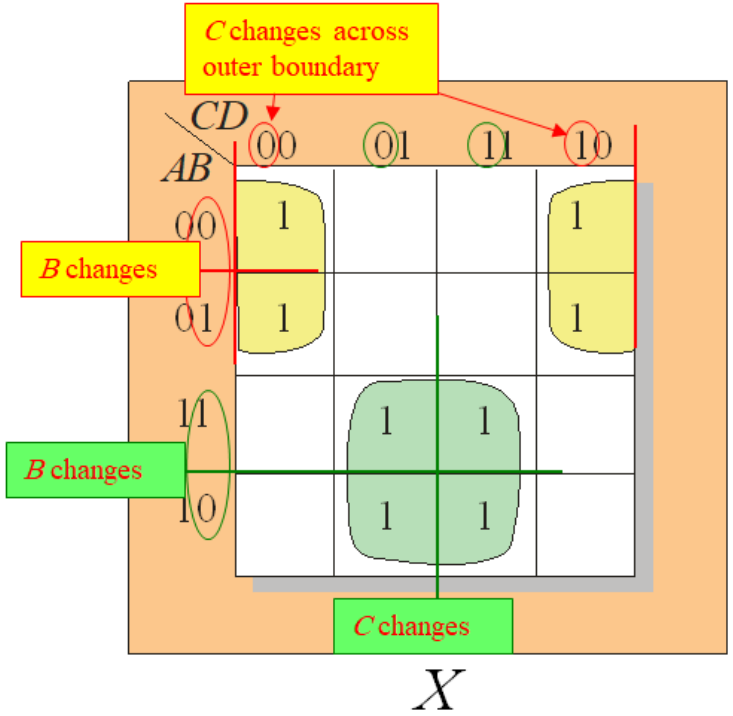
# 3-Variables Simplification method




1. Group the 1's into two overlapping groups as indicated.
2. Read each group by eliminating any variable that changes across a boundary.
3. The vertical group is read $\overline{A}\,\overline{C}$.
4. The horizontal group is read $\overline{A}B$.

$$X = \overline{A}\,\overline{C} + \overline{A}B$$

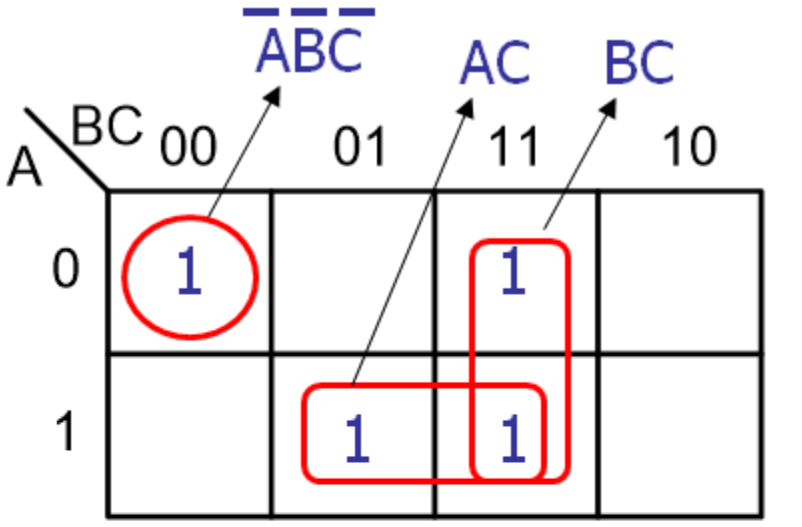
# Simplification method (Cont.)

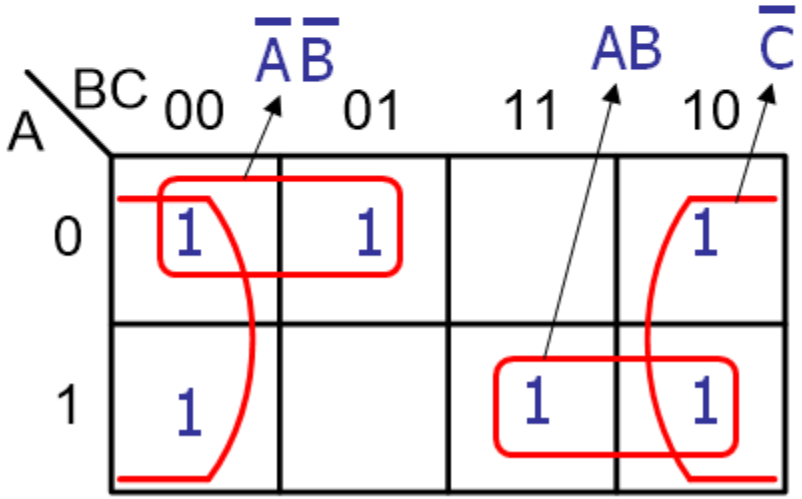


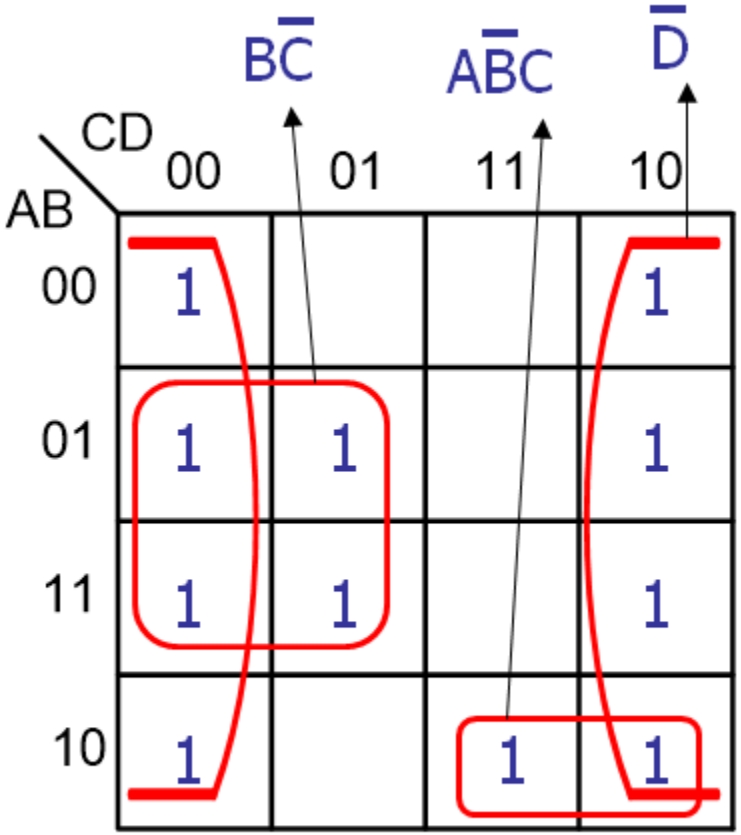

1. Group the 1's into two separate groups as indicated.
2. Read each group by eliminating any variable that changes across a boundary.
3. The upper (yellow) group is read as $\overline{A}\,\overline{D}$.
4. The lower (green) group is read as $AD$.

$$X = \overline{A}\,\overline{D} + AD$$

# Simplification method (Cont.)



# Exercise 1

Map and minimize the following std. SOP expression on a Karnaugh map: $\bar{A}\bar{B}\bar{C}+\bar{A}\bar{B}C+AB\bar{C}+A\bar{B}\bar{C}$

000 001 110 100

| A \ BC | 00 | 01 | 11 | 10 |
|---|---|---|---|---|
| 0 | 1 | 1 | | |
| 1 | 1 | | | 1 |

$\bar{A}\bar{B}$

| A \ BC | 00 | 01 | 11 | 10 |
|---|---|---|---|---|
| 0 | 1 | 1 | | |
| 1 | 1 | | | 1 |

$A\bar{C}$

Answer: $\bar{A}\bar{B}+A\bar{C}$

# Exercise 2

Map and minimize the following SOP expression on a Karnaugh map: $AB + \bar{A}\bar{B}\bar{C} + \bar{A}BC$

110 111 010 011

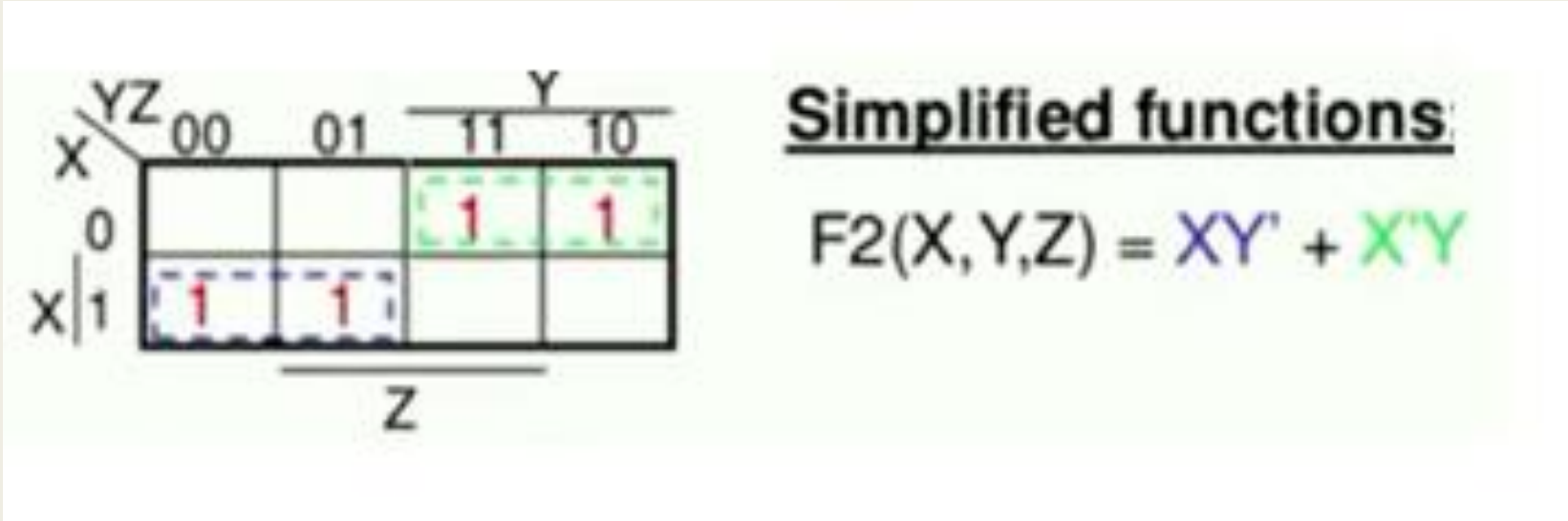
| A \ BC | 00 | 01 | 11 | 10 |
|---|---|---|---|---|
| 0 | | | 1 | 1 |
| 1 | | | 1 | 1 |

| A \ BC | 00 | 01 | 11 | 10 |
|---|---|---|---|---|
| 0 | | | 1 | 1 |
| 1 | | | 1 | 1 |

B

Answer: B

# 3-Variable K-Map Example simp.



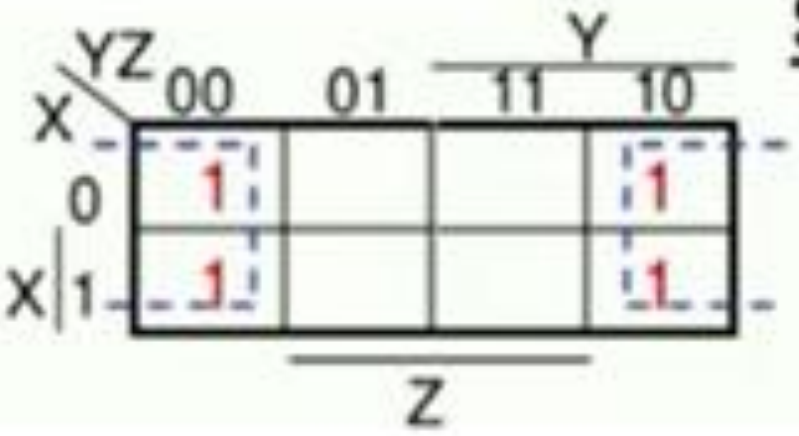

X’YZ + X’YZ’ + XY’Z’ + XY’Z

X’Y (Z+Z’) + XY’(Z’+Z)

X’Y (1) + XY’(1)

X’Y + XY’

# 3-Variable K-Map Example 2




**Simplified functions**

F3(X,Y,Z) = Z'

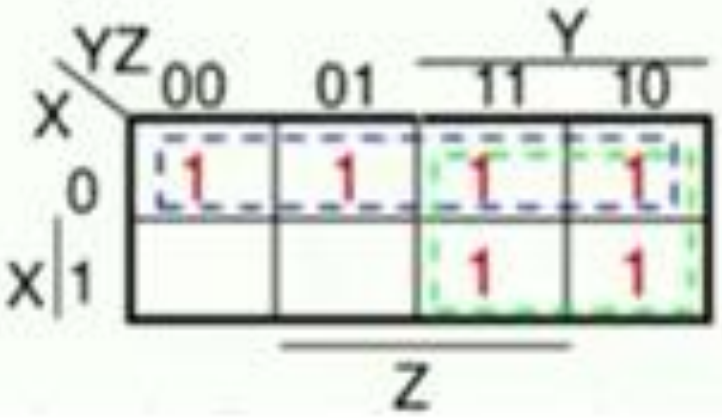
F3(X,Y,Z) = Y'Z' + Y Z'

= Z'(Y+Y')

= Z'

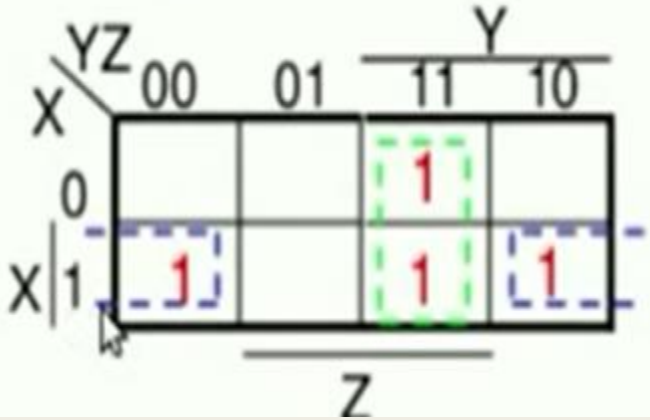
# 3-Variable K-Map Example 3

**Simplified functions:**




$F4(X,Y,Z) = X' + Y$

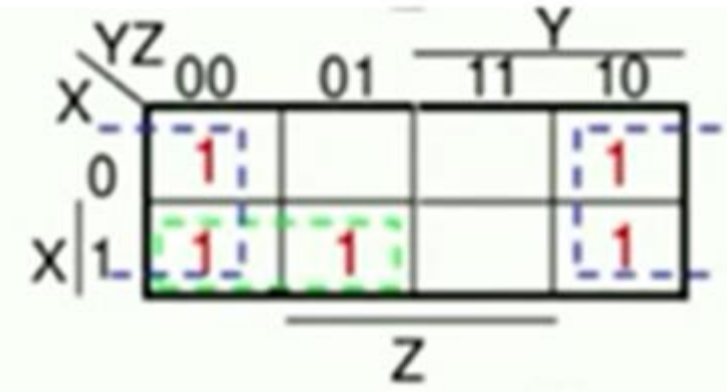
# 3-Variable K-Map Example 4




**Simplified functions**

$F5(X,Y,Z) = XZ' + YZ$

# 3-Variable K-Map Example 5




**Simplified functions:**

$F6(X,Y,Z) = Z' + XY'$

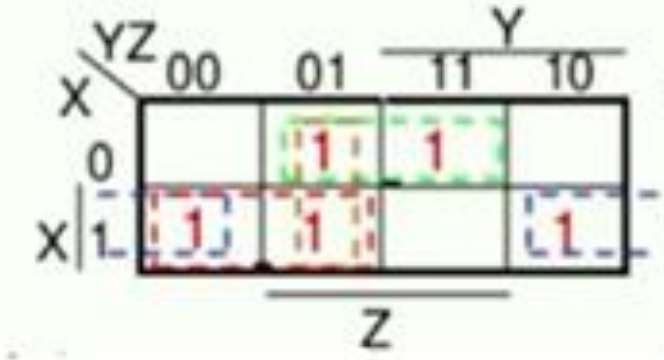
# 3-Variable K-Map Example 6

| X \ YZ | 00 | 01 | 11 | 10 |
|---|---|---|---|---|
| 0 | | 1 | 1 | 1 |
| 1 | | 1 | 1 | |

**Simplified functions**

$$F7(X,Y,Z) = Z + X'Y$$

# 3-Variable K-Map Example 7




**Simplified functions**

$F8(X,Y,Z) = XZ' + X'Z + Y'Z$

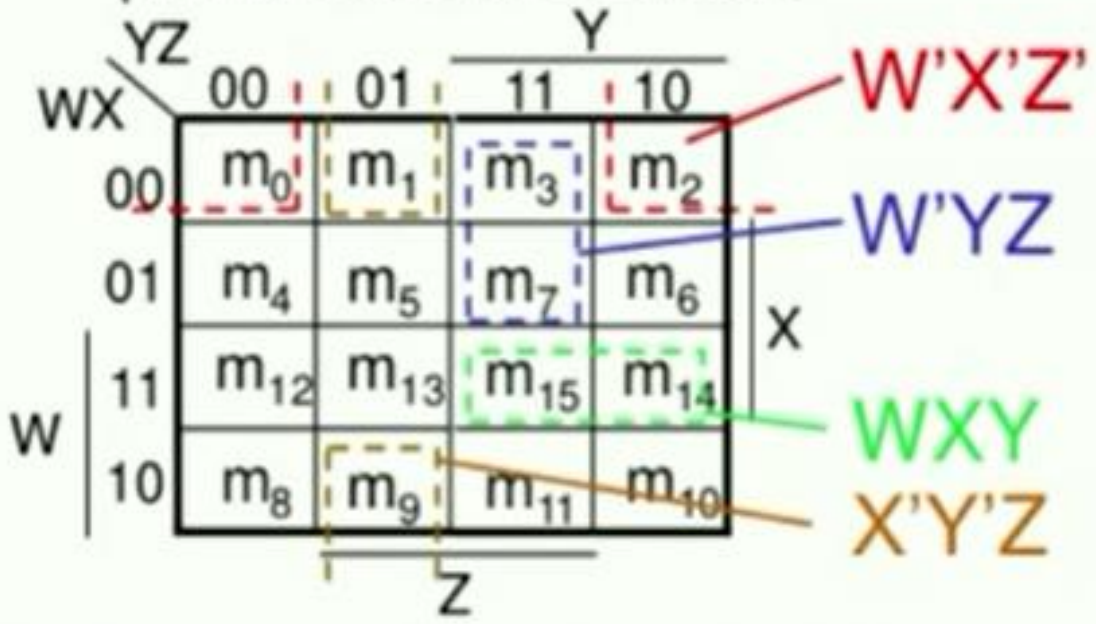
or

$F8(X,Y,Z) = XZ' + X'Z + XY'$

Not unique solution

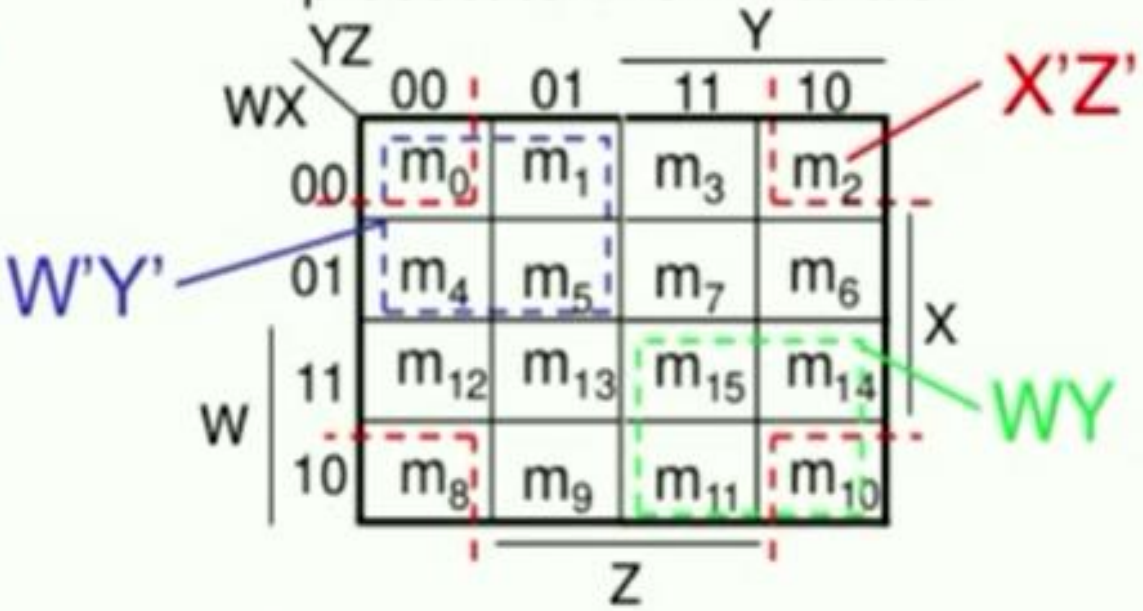
# 4-Variable K-Map Example 1

- Group of 2 adjacent cells gives product term of 3 literals.



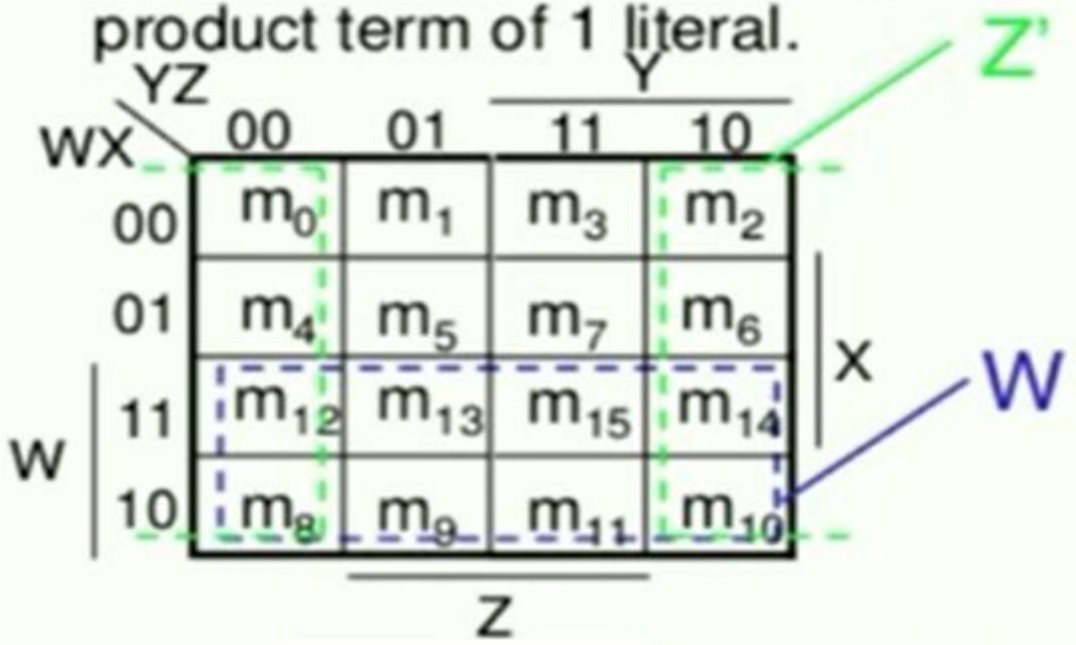
# 4-Variable K-Map Example 2

- Group of 4 adjacent cells gives product term of 2 literals.



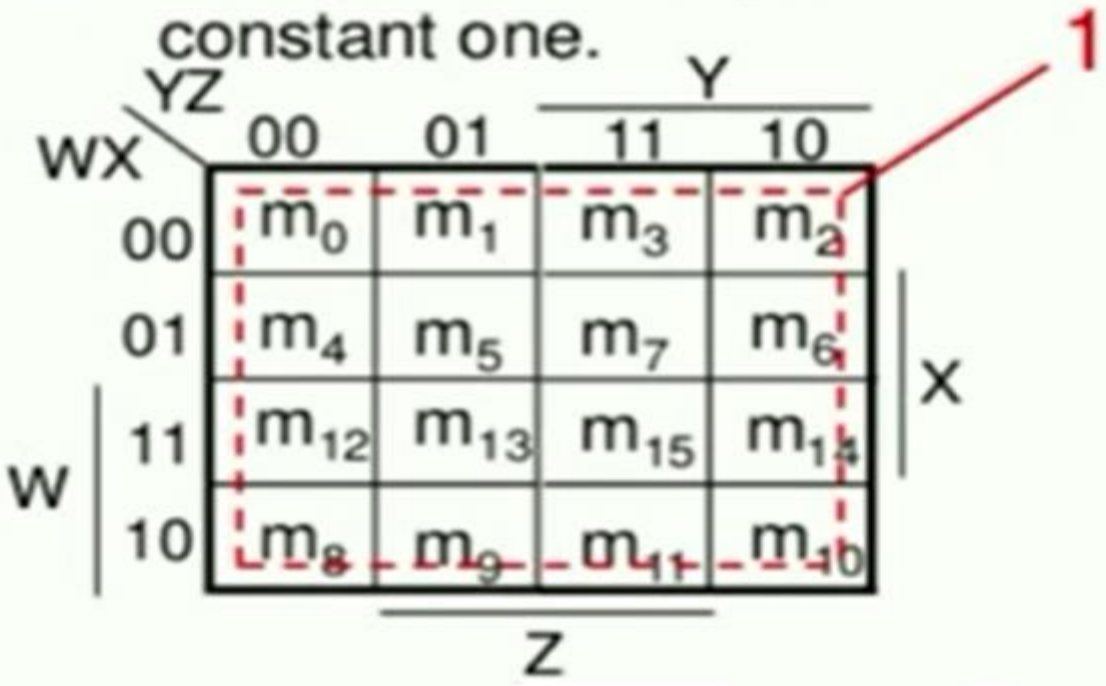
# 4-Variable K-Map Example 3

- Group of 8 adjacent cells gives product term of 1 literal.



# 4-Variable K-Map Example 4

- Group of all cells gives constant one.



# 4-Variable K-Map Example 5

| WX \ YZ | 00 | 01 | 11 | 10 |
|---|---|---|---|---|
| 00 | 1 | 1 | | 1 |
| 01 | 1 | 1 | 1 | |
| 11 | 1 | 1 | | |
| 10 | 1 | 1 | | 1 |

**Simplified function:**

F1(W,X,Y,Z) =

$= Y' + X'Z' + W'XZ$

# 4-Variable K-Map Example 6

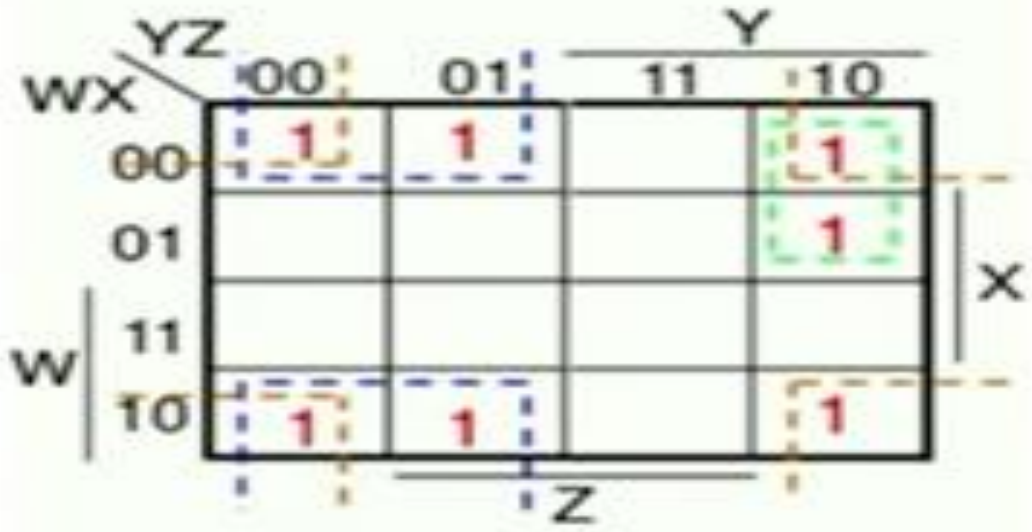
| WX \ YZ | 00 | 01 | 11 | 10 |
|---|---|---|---|---|
| 00 | 1 | 1 | | 1 |
| 01 | 1 | 1 | | 1 |
| 11 | 1 | 1 | | 1 |
| 10 | 1 | 1 | | |

**Simplified function:**

F2(W,X,Y,Z) =

= Y' + W'Z' + XZ'

# 4-Variable K-Map Example 7




**Simplified function:**

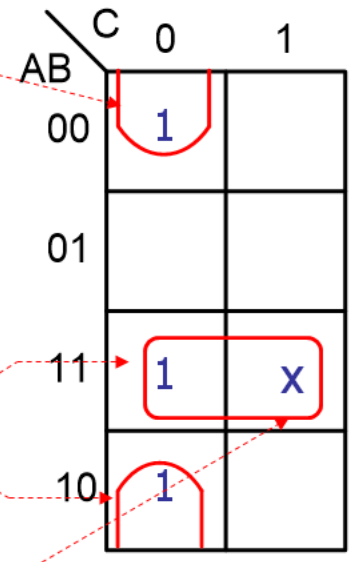
F3(W,X,Y,Z) =

= X'Y' + X'Z' + W'YZ'

# From truth table to K-Map

## Mapping Directly from a Truth Table

| A | B | C | Out |
|---|---|---|---|
| 0 | 0 | 0 | 1 |
| 0 | 0 | 1 | 0 |
| 0 | 1 | 0 | 0 |
| 0 | 1 | 1 | 0 |
| 1 | 0 | 0 | 1 |
| 1 | 0 | 1 | 0 |
| 1 | 1 | 0 | 1 |
| 1 | 1 | 1 | x |




$Out = AB + \overline{B}\,\overline{C}$

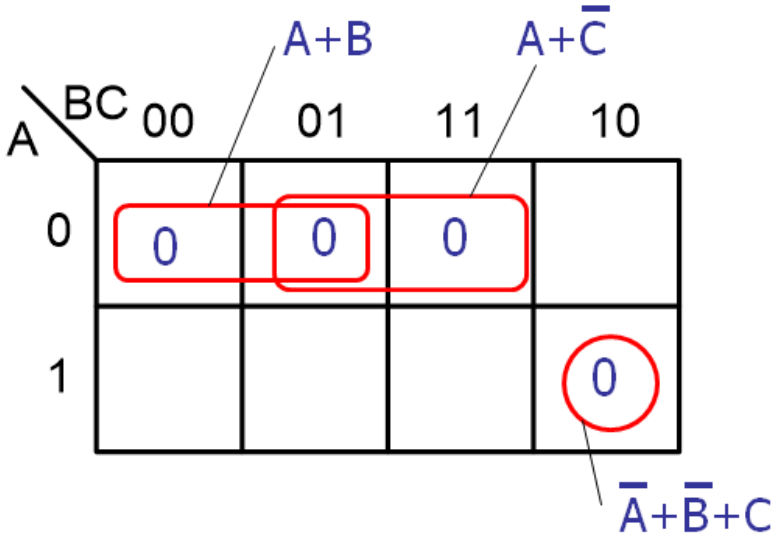
# Mapping POS (Zeros)

Map and minimize the following std. POS expression on a Karnaugh map:

$(A+B+C)(A+B+\bar{C})(\bar{A}+\bar{B}+\bar{C})(\bar{A}+\bar{B}+C)$

000 001 111 110

| A \ BC | 00 | 01 | 11 | 10 |
|---|---|---|---|---|
| 0 | 0 | 0 | 0 | |
| 1 | | | | 0 |




Answer: $(A+B)(A+\bar{C})(\bar{A}+\bar{B}+C)$

Prime = 1

# Mapping POS (cont.)

Map and minimize the following POS expression on a Karnaugh map:

$(A+B)(A+\overline{B}+C)(A+\overline{B}+\overline{C})$

000 001 010 011

| A \ BC | 00 | 01 | 11 | 10 |
|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 |
| 1 | | | | |

A

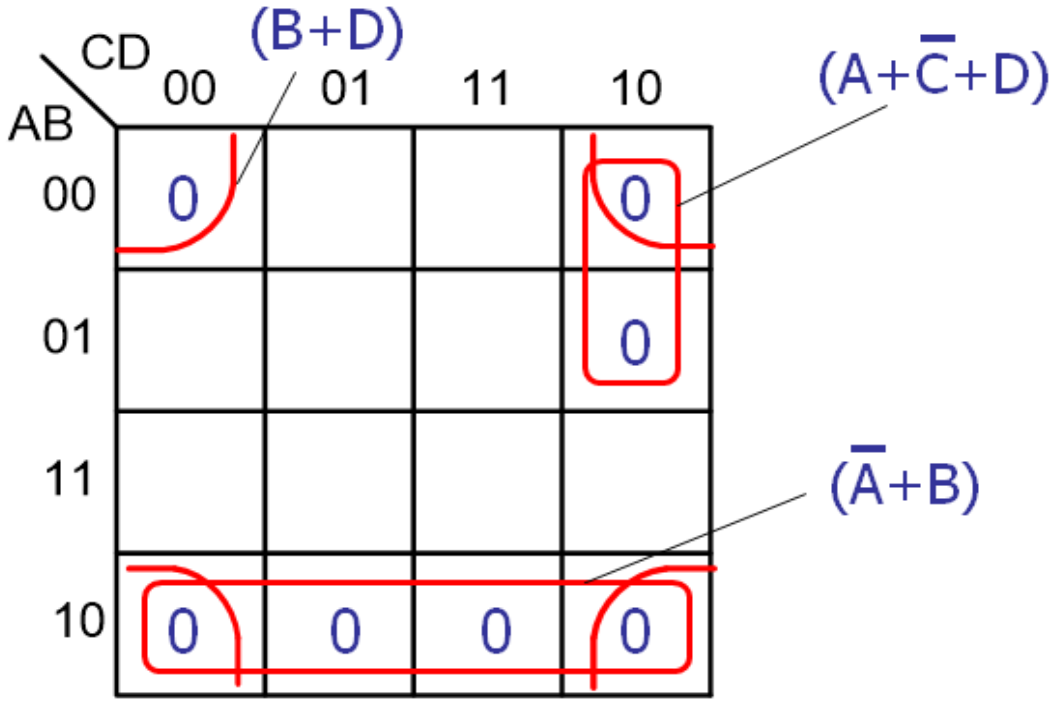
| A \ BC | 00 | 01 | 11 | 10 |
|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 |
| 1 | | | | |

Answer: A

# Mapping POS (Cont.)

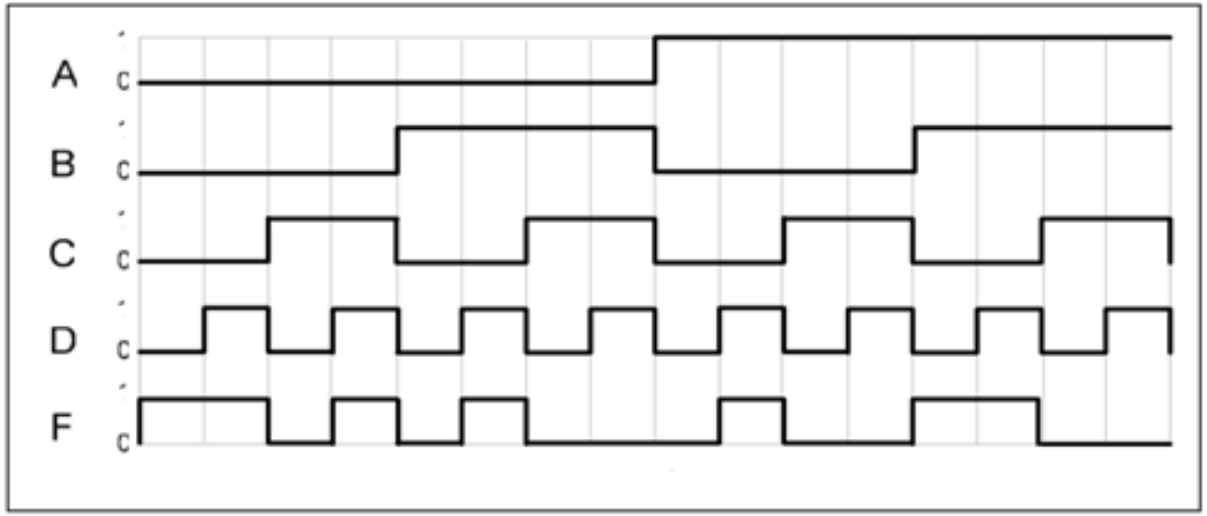
$(B+C+D)(A+\overline{C}+D)(\overline{A}+B+\overline{C}+\overline{D})(\overline{A}+B+C+\overline{D})(\overline{A}+B+\overline{C}+D)$

0000 1000 0010 0110 1011 1001 1010




Answer: $(B+D)(\overline{A}+B)(A+\overline{C}+D)$

# Experiment 1




For the Timing Diagram given in the above figure:

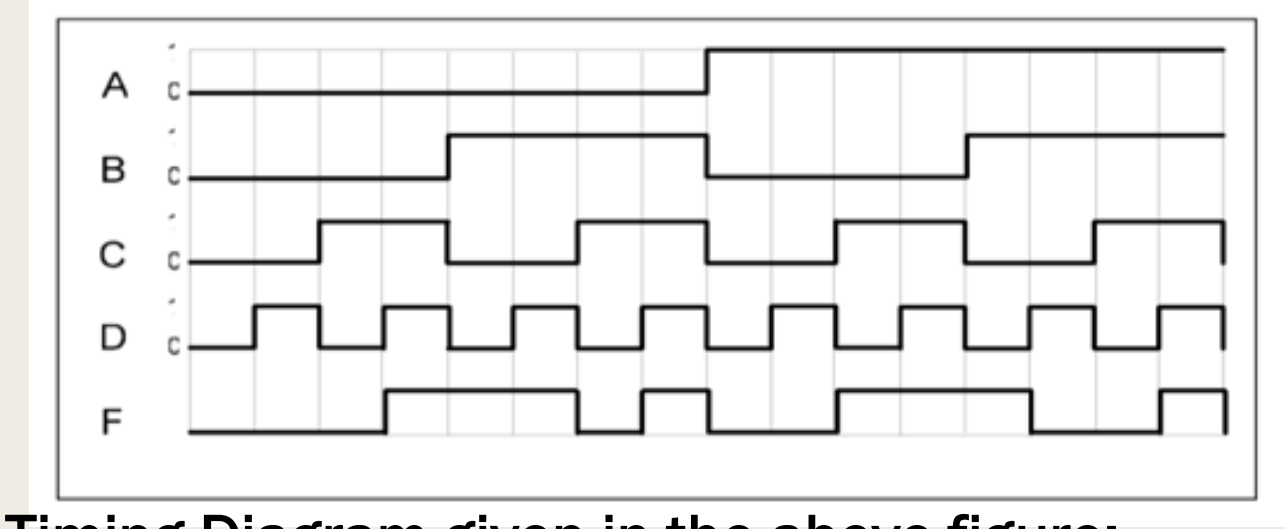
1. Obtain the Truth table of F from the timing diagram

2. Obtain the K-Map from the truth table

3. Express the function F as:
   - Sum of Minterms expression (SOP)
   - Product of Maxterms expression (POS)

4. Draw the logic circuit diagram for F using the basic gates

# Experiment 2




For the Timing Diagram given in the above figure:

1. Obtain the Truth table of F from the timing diagram

2. Obtain the K-Map from the truth table

3. Express the function F as:
   - Sum of Minterms expression (SOP)
   - Product of Maxterms expression (POS)

4. Draw the logic circuit diagram for F using the basic gates

# Experiment 3

Design a circuit with output G and inputs $x_1$, $x_0$, $y_1$, and $y_0$. Let $X = x_1x_0$ be a number, where the four possible values of X, namely, 00, 01, 10, and 11, represent the four numbers 0, 1, 2, and 3, respectively. Similarly, let $Y = y_1y_0$ represent another number with the same four possible values. The output G should be 1 only if the number X is greater than or equal to the number Y.

1. Obtain the truth table for G.
2. Express function G as a Sum of Minterms expression.
3. Express function G as Product of Maxterms expression.
4. Find the minimum-cost SOP and POS forms for G using the map method.
5. Draw the logic diagram for G using the basic gates.

**Remind:**

| Maxterm | Minterm |
|---|---|
| • F = 0 | • F= 1 |
| • (+ +).(+ +).(+ +) | • (. .) +(. .) +(. . ) |
| • 1 → prime | • 0 → prime |

| X | | Y | | G |
|---|---|---|---|---|
| X1 | X0 | Y1 | Y0 | |
| 0 | 0 | 0 | 0 | |
| 0 | 0 | 0 | 1 | |
| 0 | 0 | 1 | 0 | |
| 0 | 0 | 1 | 1 | |
| 0 | 1 | 0 | 0 | |
| 0 | 1 | 0 | 1 | |
| 0 | 1 | 1 | 0 | |
| 0 | 1 | 1 | 1 | |
| 1 | 0 | 0 | 0 | |
| 1 | 0 | 0 | 1 | |
| 1 | 0 | 1 | 0 | |
| 1 | 0 | 1 | 1 | |
| 1 | 1 | 0 | 0 | |
| 1 | 1 | 0 | 1 | |
| 1 | 1 | 1 | 0 | |
| 1 | 1 | 1 | 1 | |

Computing G:

X > = Y → G = 1

X < y → G = 0

# Homework

Given the following Truth table:

| A | B | C | F1 | F2 | F3 | F |
|---|---|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 1 | 0 | 1 |
| 0 | 0 | 1 | 1 | 0 | 1 | 0 |
| 0 | 1 | 0 | 1 | 1 | 0 | 1 |
| 0 | 1 | 1 | 0 | 0 | 1 | 1 |
| 1 | 0 | 0 | 0 | 1 | 0 | 1 |
| 1 | 0 | 1 | 1 | 0 | 1 | 0 |
| 1 | 1 | 0 | 1 | 1 | 0 | 1 |
| 1 | 1 | 1 | 0 | 0 | 1 | 1 |

1. Express the functions F1, F2, F3 and F as a Sum of Minterms expression
2. Express the functions F1, F2, F3 and F as a Product of Maxterms expression
3. Simplify the Function F using the map method
4. Draw and simulate the digital circuit diagram for F using the basic gates